Data Source Report: Wine Quality Dataset
1. Overview & Characteristics

Field	Detail
Dataset Name	Wine Quality Dataset
Source	UCI Machine Learning Repository
Dataset Link	Wine Quality Dataset (ID: 186)
Description	Contains physicochemical properties of red and white Portuguese wines and their quality ratings.
Data Type	Secondary Data
Why Secondary?	The data was already collected and published by researchers. It was not independently gathered by the author.
Format	CSV / Tabular Data

2. Volume

- **Red Wine:** 1,599 records
- **White Wine:** 4,898 records

3. Features

Input Features (Physicochemical)	Target Feature (Sensory)
Fixed Acidity	Quality
Volatile Acidity	
Citric Acid	

Residual Sugar	
Chlorides	
Free Sulfur Dioxide	
Total Sulfur Dioxide	
Density	
pH	
Sulphates	
Alcohol	

4. Purpose & Reliability

- **Purpose:** To analyze factors affecting wine quality and build predictive analytics models.
- **Reliability:** The dataset is hosted by the UCI Machine Learning Repository, a highly trusted academic source widely used in research and machine learning projects globally.

5. Sample Raw Data

Python Code:

```
from ucimlrepo import fetch_ucirepo

wine_quality = fetch_ucirepo(id=186)

X = wine_quality.data.features
y = wine_quality.data.targets

print("Dataset Shape:")
print(X.shape)

print("\nFirst 5 Rows:")
print(X.head())

print("\nTarget Values:")
print(y.head())
```

```
print("\nColumn Names:")
print(X.columns)
```

OUTPUT : -

```
Dataset Shape:
(6497, 11)

First 5 Rows:
   fixed_acidity  volatile_acidity  citric_acid  residual_sugar  chlorides  ...  total_sulfur_dioxide  density    pH  sulphates  alcohol
0            7.4              0.70         0.00             1.9      0.076  ...                  34.0   0.9978  3.51       0.56      9.4
1            7.8              0.88         0.00             2.6      0.098  ...                  67.0   0.9968  3.20       0.68      9.8
2            7.8              0.76         0.04             2.3      0.092  ...                  54.0   0.9970  3.26       0.65      9.8
3           11.2              0.28         0.56             1.9      0.075  ...                  60.0   0.9980  3.16       0.58      9.8
4            7.4              0.70         0.00             1.9      0.076  ...                  34.0   0.9978  3.51       0.56      9.4

[5 rows x 11 columns]

Target Values:
   quality
0        5
1        5
2        5
3        6
4        5

Column Names:
Index(['fixed_acidity', 'volatile_acidity', 'citric_acid', 'residual_sugar',
       'chlorides', 'free_sulfur_dioxide', 'total_sulfur_dioxide', 'density',
       'pH', 'sulphates', 'alcohol'],
      dtype='object')
PS E:\Data_Analytics>
```